Prakrati Gupta

+91 8077021259 | prakratigupta76@gmail.com | linkedin.com/in/prakratigupta76 | github.com/prakratigupta76

Summary

Data Analyst and Machine Learning enthusiast with an M.Sc. in Statistics and hands-on experience in **Python, SQL, Power BI, and VBA** across EDA, predictive modeling, and process automation. Currently an **Associate Analyst at Infosys BPM**, building **AI-driven automation solutions** using **Python and Azure OpenAI**. Comfortable turning messy or loosely-defined problems into clear, data-backed answers. Looking for a data analyst, data scientist, or ML/AI role where strong statistical thinking meets practical modeling and automation skills.

Technical Skills

Programming Languages: Python, SQL, Core Java, VBA
Data Analysis & Visualization: Excel, Power BI, SPSS, Matplotlib, Seaborn, Power Query
Machine Learning: Scikit-learn, XGBoost, Random Forest, Support Vector Machine, Logistic Regression, Linear Regression, Decision Trees
Data Science: Exploratory Data Analysis (EDA), Feature Engineering, Predictive Modeling, Statistical Analysis, Hypothesis Testing, Forecasting
Libraries & Tools: Pandas, NumPy, Streamlit, ETL Pipelines, Azure OpenAI
Soft Skills: Analytical Thinking, Problem Solving, Team Collaboration, Stakeholder Communication

Experience

Associate Analyst Aug 2026 – Present
Infosys BPM

- Working within the **AI & Automation** function, applying **Python, SQL, and VBA** to build automation solutions and support data-driven workflows.
- Currently developing an **AI & Automation tool** using **Python and Azure OpenAI**, integrated with **SQL** for data handling, aimed at improving process efficiency and reducing manual effort.
- Automating manual, repetitive business processes using **Python** to reduce turnaround time and cut down on manual effort.
- Writing and optimizing **SQL queries** to extract, transform, and report on business data in support of ongoing automation and AI initiatives.

Artificial Intelligence Intern Jul 2025 – Jul 2026
Infosys BPM

- Designing and validating **AI/ML models** in **Python** for real-world business use cases — models that feed into actual workflows, not just experiments.
- **Data preprocessing, feature engineering, model training, and evaluation** are all handled end-to-end; improving model accuracy and reliability is the constant goal.
- Regular collaboration with cross-functional teams to debug, optimize, and deploy **machine learning solutions** — this is where a lot of the practical problem-solving happens.
- Develop and run **test cases and validation processes** to catch performance issues before models go live.
- Upskilling through structured programs in **machine learning, data analysis**, and software development best practices — running alongside live project work.

Projects

AI-Based Real Estate Valuation System | *Python, XGBoost, Scikit-learn, Streamlit, Pandas, NumPy* GitHub

- End-to-end **supervised ML pipeline** built on the Ames Iowa Housing Dataset — **1,460 records and 80 features** — to predict residential property prices.
- Created **8 domain-specific features** (TotalSF, HouseAge, TotalBath, HasPool, etc.); log-transformed the target variable to handle skewness, both of which improved model fit noticeably.
- Trained and benchmarked Linear Regression, Decision Tree, Random Forest, and **XGBoost** across RMSE, MAE, and R^2 — **XGBoost** delivered the best result at $\mathbf{R^2 = 0.905}$ and **RMSE ~ $23,000**.
- Deployed a **Streamlit web app** with three tabs: live price predictor with **confidence intervals**, model insights with **feature importance charts**, and an interactive market trends explorer.

Predictive Modeling – Car Price Forecasting | *Python, Scikit-learn, Matplotlib, Seaborn* GitHub

- **Exploratory data analysis** on **50,000+ records** to identify the features that genuinely drive car prices — not all the obvious ones held up.
- Compared **Linear Regression, Random Forest, and SVM**; **feature engineering and hyperparameter tuning** pushed prediction accuracy up by **18%**.
- Visualized **feature importance** and price trends using **Matplotlib and Seaborn** to present findings clearly to stakeholders.

Heart Attack Risk Prediction – Classification Model | *Python, Pandas, NumPy, Scikit-learn* GitHub

- **Logistic Regression** classification model for heart attack risk prediction — **AUC score of 0.822**, which holds up well for a clinical screening task.
- Used **Chi-square and ANOVA hypothesis testing** to identify statistically significant risk factors before modeling, not after.
- **Feature selection and data preprocessing** kept the model interpretable — important here because the output informs healthcare decisions, not just a dashboard.

iEMS – Infosys Email Management System | *Python, PySide6, PyQt, pyodbc, SQL Server* *Internal Project*

- Collaborated in a team to build **iEMS**, a desktop application (**Python, PySide6/PyQt**) that automates email allocation, tracking, and reporting for teams handling high email volume.
- Independently developed the **Clarification, Reports, and My Profile** modules end-to-end – designing the front-end UI in **PySide6/PyQt** and the backend logic connecting to **SQL Server** via **pyodbc**.
- Used **pywin32** for Outlook/Windows integration and **PyInstaller** to package the application for deployment; used **Pandas and NumPy** for report generation and data processing.

Education

M.Sc. in Statistics 2021 – 2023
Dr. Bhimrao Ambedkar University, Agra

B.Sc. in Physics, Statistics & Mathematics (PSM) 2018 – 2021
Dr. Bhimrao Ambedkar University, Agra

Certifications

Microsoft Power BI Data Analyst Professional Certificate
IBM Data Visualization and Dashboards (Excel, Cognos)
Complete Data Science Bootcamp (Python, Machine Learning) – Udemy
Explore Machine Learning Using Python – Infosys Springboard (Python, Machine Learning, Data Analysis)